

GALAXIES

GALAXIES ARE FOUND IN A HUGE VARIETY OF SHAPES AND SIZES, FROM COMPLEX SPIRALS LIKE OUR MILKY WAY TO HUGE BALLS OF ANCIENT RED AND YELLOW STARS, AND SHAPELESS CLOUDS OF GAS, DUST, AND NEWBORN STARS.

Galaxies are the only places in the Universe where matter is densely packed enough for stars to form, and most stars spend their whole lives within them. Held together by gravity, most galaxies are thought to have a supermassive black hole at their center.

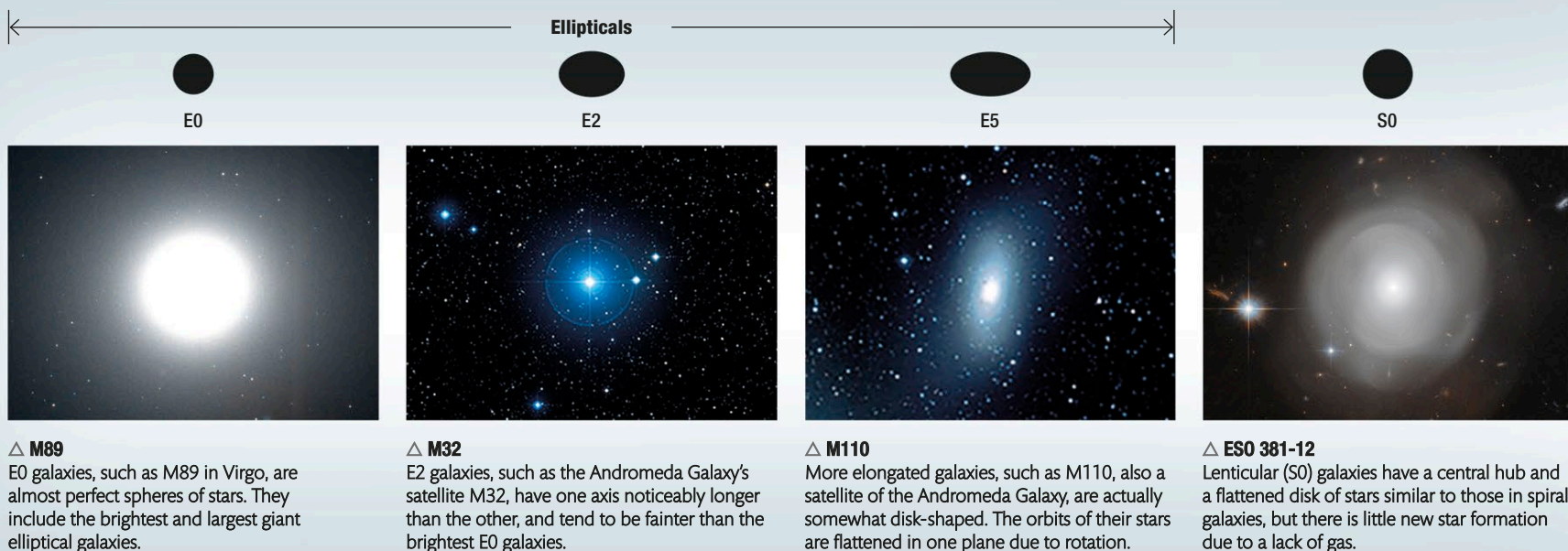
Types of galaxies

American astronomer Edwin Hubble confirmed the existence of galaxies beyond the Milky Way in the 1920s. He subdivided them into several distinct types distinguished by a code of letters and numbers. Elliptical galaxies (types E0 to E7) are all roughly ball-shaped, but range from rounded spheres to elongated cigars. Today we know that they are dominated by old red and yellow stars. Spirals (types S and SB) are flattened disks with dense areas of star formation in the spiral arms, and older red and yellow stars in the center. Lenticulars (type S0) have a central hub surrounded by a disk, but no spiral arms, while Irregulars (type Irr I and II) are fairly shapeless clouds rich in star-forming material.



△ Elliptical galaxies

Elliptical galaxies, such as M60 (shown here with the spiral galaxy NGC 4647), are ball-shaped star systems created by countless stars in overlapping elliptical orbits tilted at a wide range of angles. They have very little of the gas needed to support new star formation, which leaves them dominated by long-lived, low-mass red and yellow stars. They range in size from sparsely populated dwarfs to vast “giant ellipticals,” which are the largest galaxies in the Universe.



△ Hubble's tuning fork

Edwin Hubble arranged the various galaxy types in the shape of a musical tuning fork. He thought this illustrated the way that galaxies evolve over time, although the true story is rather more complex (see pp.62–63). Ellipticals are numbered according to their shape, with E0 the most round in shape. There are two distinct types of spiral galaxy—normal spirals (type Sc to Sc or Sd), in which the spiral arms emerge directly from the central hub, and barred spirals (SBa to SBc), in which the arms attach to the ends of a straight bar crossing the hub.

Astronomers think there are as many galaxies in our Universe as there are stars in the Milky Way Galaxy